

● Marketing analytics case study · Google Data Analytics capstone

# Where should the Google Merchandise Store spend its next marketing dollar?

Channel attribution, a retargeting model and two tests that would settle what attribution can't

Google Analytics 360 export

Aug 2016 – Aug 2017

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Analysis designed and directed by me; code written with AI assistance.



## Five findings for next quarter's budget

**Up to about 15**

points of credit GA's report likely gives Organic Search that return visits earned

**34%**

of purchases come from visitors returning on their own (44% of capped revenue); when they first came from a campaign, GA's report credits that campaign

**\$0.78–1.21**

is the most a Paid Search click is worth at a 50% margin, if every sale needed the ad

**71%**

of later buyers are in the model's top 10%, yet retargeting them is worth only up to about \$690–745 a month before ad costs

**89%**

of the revenue GA credits to Display came from one corporate buyer: hold Display's budget and test it on site visits

● ● Before any analysis

# 41%

## of the store's revenue came from Google employees, not marketing

In the public data these visits appear as "Referral" with the source hidden. Google's training copy shows the source, [mall.googleplex.com](https://mall.googleplex.com), the employee store link.

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### 5%

of visitors

### 47%

of purchase sessions

### 99.2%

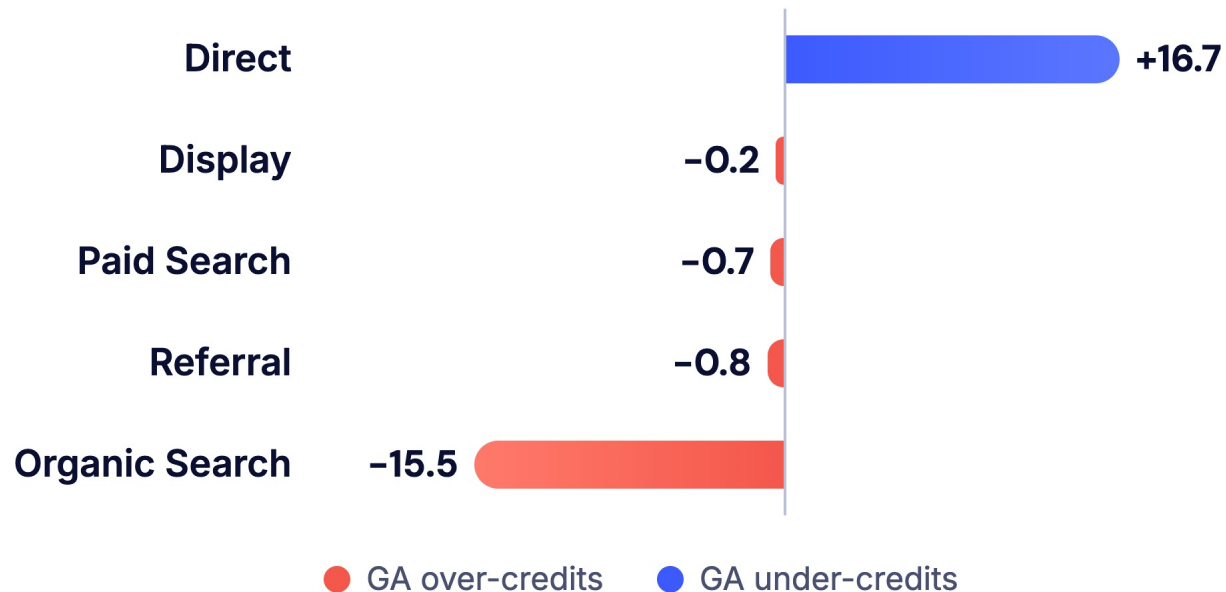
precision of my employee  
filter against that copy

### 98.2%

of employee purchases the  
filter catches

# GA likely over-credits Organic Search by up to about 15 points

Share of purchases a data-driven model credits, minus the share GA's report credits, in points, if every direct return was self-initiated



## How big is the gap?

**15.5**

points if every direct return was self-initiated  
(95% CI 14.4–16.6)

**13–15**

points at a benchmark from how returning  
visitors behave

**1.8**

points on GA's own labels

●● Channel budget: what a paid click is worth

# A Paid Search click's value is a ceiling, not a profit

Bid cap per Paid Search click at a 50% gross margin

| If the ads caused...  | Bid cap per click |
|-----------------------|-------------------|
| every attributed sale | \$0.78–\$1.21     |
| half of them          | \$0.39–\$0.61     |
| a quarter of them     | \$0.19–\$0.30     |

A click is credited with \$1.56–\$2.43 of revenue under seven attribution rules; the cap is half of that times the share the ads caused.

## 65 of 65

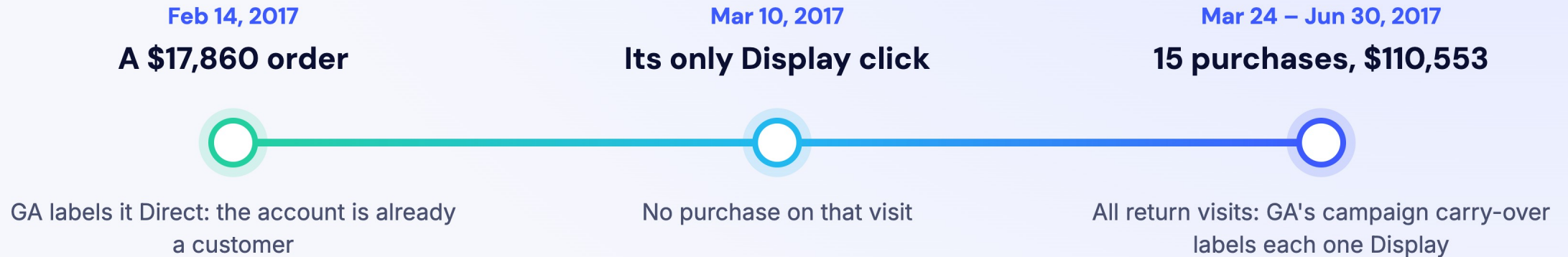
Paid Search purchases with a readable keyword came on brand keywords

## 231 of 437

buying journeys with Paid Search touch no other channel, so five of the seven rules give it full credit

➔ Split brand from non-brand campaigns, and test pausing brand ads (not yet sized), before raising any bid.

# One corporate buyer made Display look like a winner



**89%**

of the revenue GA credits to Display is this one account

**\$2.84–\$3.87**

a Display click's attributed value without it, under every rule

## Recommendation

**Report the account on its own, like employees, and test Display on site visits.**

# The model's top 10% holds 71% of later buyers



Later buyers reached by targeting the top 10%

**My model (gradient boosting)**



**Two-line rule: North America, then funnel step**



**Random targeting**

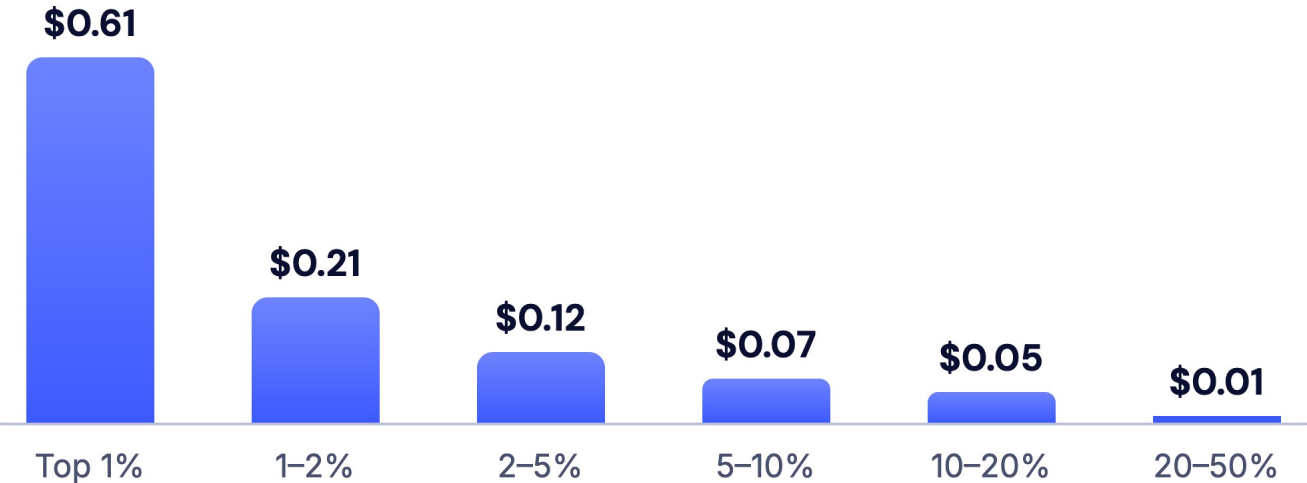


**What predicts a return purchase (training months)**

A North American first visit (**15.7×** the odds),  
adding to cart (**3.7×**), and arriving by  
Paid Search or Display (about **15–21×** Social's odds)

# Only top-scored visitors are worth paying to bring back

The most worth paying per visitor the ads actually reach, for 30 days, by score band



Staff flagged using the whole year; without hindsight, each band moves a few cents

**\$690–745**

a month at most in extra gross profit from the top 10%, if the ads reach everyone (before ad costs)

**\$920–950**

a month at most for the top 20%

**\$207–\$1,936**

sensitivity range: the top 10% a month at most, at 5–20% lift and 30–70% margin

## Eight actions: four now, two tests, two to explore

### Now

**Report employees and the key account as their own segments**

Web analytics  
41% of revenue; \$128k

**Show a multi-touch view beside GA's channel report**

Head of Marketing, Finance  
up to about 15 pts of credit

**Split brand from non-brand search; bid under the ceiling**

Paid media  
\$1.56–2.43 per click

**Review any spend on YouTube promotion and affiliates**

Paid media  
Oct–Nov: 98k visits, no sales

### Test

**Retarget only top-scored visitors; 50/50 holdout test for 12 months**

Retargeting lead  
Up to about \$690–950  
a month (top 10–20%)

**Hold Display's budget; run a 12-week site-visit holdout**

Paid media  
\$17k revenue a year

### Explore

**Retention: email and reminders for past visitors**

Head of Marketing  
34% of purchases

**A direct sales path for corporate buyers**

Head of Marketing, Sales  
\$249k bulk purchases

## Four decisions for today

**1 Report employees and the key account on their own?**

Then every channel number matches this analysis. Employees alone are 41% of revenue.

**2 Run a 12-month, 50/50 retargeting test of the top 20%?**

The only design that can measure the lift: it detects a lift of 14% or more; 10% would need about 23 months.

**3 Hold Display's budget and test its own audience on site visits for 12 weeks?**

About 2 purchases a week are too few to test. Visits work if that audience makes at most about 500–1,000 visits a week.

**4 Split brand from non-brand search and test pausing brand ads (not yet sized)?**

Every readable keyword behind a Paid Search purchase was a brand keyword (the store's or Google's name).

# How sure I am

## ● High confidence

Employee traffic, checked against Google's unredacted data

The key account's timeline, from its own visits

The retargeting ranking, tested on unseen months: 7–10 points ahead of the two-line rule

## ● Moderate confidence

Organic Search's over-credit: the direction is clear, the size rests on an assumption

The returning-visitor share (34%), for the same reason

Retargeting's value: it assumes a 10% lift, a 50% margin and full reach

## ● Low confidence

Whether Paid Search clicks are incremental: every purchase with a readable keyword came on a brand keyword

What Display adds

The retargeting lift, borrowed from published experiments

→ Attribution describes paths, not causes. The recommended tests are designed to settle the rest.

## The teaching lab's model mostly finds employees

Share of first-time visitors who are Google employees, by the lab model's predicted chance of a later purchase (May–Jun 2017)



**61%**

of the future buyers the lab model learns from are Google employees

**2 of 1,020**

top-ranked visitors are outside customers who later bought

# On outside visitors, the lab model scores 0.86, not 0.91

ROC-AUC of BigQuery ML models on the lab's evaluation months, May–Jun 2017

| Model                              | All first visits | Outside visitors |
|------------------------------------|------------------|------------------|
| The lab's model, as published      | 0.910            | 0.863            |
| Without source, medium and channel | 0.885            | 0.882            |
| Trained on outside visitors only   | —                | 0.877            |

ML.EVALUATE reports 0.909 for the recreated model (published 0.91); recomputed from row-level predictions it is 0.910.

–0.047

drop in its ROC-AUC on outside visitors  
(95% CI -0.059 to -0.035)

59–66%

of the top 1% are still employees in  
both audit versions, through geography  
and engagement



Fine for teaching; take internal traffic out before using a model like this to target anyone.

# How I did it

## Data and SQL

903,653 sessions queried in BigQuery, with nested page hits unpacked for funnel steps

Google employees and one key account reported as their own segments

A keyword extract (0.25 GB) to split brand from non-brand search

## Models and tests

Chi-square, z-tests and logistic regression with odds ratios

Logistic regression, random forest and gradient boosting, tuned on earlier months

Seven attribution rules, including a third-order Markov chain

Power analysis for the retargeting and Display tests (brand test not yet sized)

## Validation

Tested on months the models never saw in training or tuning

Bootstrap intervals that resample visitors

Over 200 automated tests; a fresh clone runs end to end

Google's lab recreated in BigQuery ML from its own SQL